

ROBIN BILODEAU

Ottawa, Ontario, Canada

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EDUCATION

University of Ottawa

Ottawa, Ontario, Canada

Bachelor of Science in Computer Science and Mathematics

Sep 2024 – Apr 2028

- **Relevant Courses:** Intro to Computing 1, Intro to Computing 2, Linear Algebra, Differential Equations and Laplace Transforms, Discrete Structures, Elementary Real Analysis, Introduction to probability

WORK EXPERIENCE

Scotiabank - Global Banking and Markets

Toronto, Ontario, Canada

Quantitative Analyst Intern

Sep 2025 – Dec 2025

- Refactored a legacy pricing process into a modular architecture supporting multiple rates derivatives, including swaps, swaptions, caps, and floors
- Optimized performance bottlenecks in core analytical functions, leading to substantial runtime improvements
- Conducted in-depth testing and investigation of pricing models to identify product-dependent discrepancies and assess model validity, and visualized discrepancy differences across products to communicate findings clearly

RESEARCH EXPERIENCE

Privacy-Preserving Zero-Knowledge Credential Protocol

Ottawa, Ontario, Canada

Undergraduate Researcher

Feb 2026 – Present

- Implementing a zero-knowledge proof system over Pedersen commitments enabling privacy-preserving equality testing on hidden attributes, combining digital signatures, hashing, and authenticated encryption

A Data-Driven Assessment of Machine Learning Models for Climate-Sensitive Wildfire Susceptibility in Alberta

Ottawa, Ontario, Canada

Undergraduate Researcher

Feb 2026 – Present

- Abstract accepted to Earth 2026 Conference
- Developing and extending the GMDH algorithm to produce an interpretable, self-organizing machine learning pipeline for wildfire susceptibility detection

Uncertainty-Aware Statistical Extensions for HDBSCAN Clustering

Ottawa, Ontario, Canada

Undergraduate Researcher

Apr 2025 – Oct 2025

- Developed a Monte Carlo sampling framework to repeatedly adjust probability rank matrices, yielding statistically stable core-distance estimates for clustering high-noise data leading to reduced sensitivity to outliers and improved cluster detection reliability
- Coordinated a fast Fourier-based deconvolution estimator to recover underlying data distributions from noise-contaminated observations, enhancing core distance estimates for robust HDBSCAN clustering

PROJECTS

RobinBook an Interactive Limit Order Book Simulation

- Built a real-time limit order book engine in FastAPI supporting LIMIT, MARKET, and FOK order types with price-time priority matching, live Yahoo Finance price feeds, and per-symbol independent book instances for TSLA, AAPL, and GOOG
- Leveraged Python's bisect module to maintain sorted bid/ask price levels, significantly improving insert and search performance over naive list operations across high-frequency order flow
- Designed and deployed five autonomous trading bots simulating distinct market participant archetypes (market maker, trend follower, institutional whale, noise trader, and an HFT bot whose core signal logic is written in C++ and called via ctypes for sub-second latency)

LEADERSHIP AND ACTIVITIES

SSC Case Study Competition - Parkinson's Disease Classification

Hamilton, Ontario, Canada

Team Member

Apr 2026 – Jun 2026

- Performed end-to-end data cleaning on a high-dimensional clinical dataset (500+ variables) from the Canadian Open Parkinson Network (C-OPN), including leakage analysis to identify and remove variables that directly encode diagnostic criteria for PD vs. atypical parkinsonism, and outcome-stratified MICE imputation to handle missing data
- Developed a random forest classifier achieving greater than 95% accuracy in distinguishing Parkinson's disease from atypical parkinsonism syndromes, using stratified k-fold cross-validation with fold splitting prior to imputation

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Native speaker), French (Native speaker)
- **Programming:** Python, R, Java, Typst, C++ (Beginner)
- **Tools:** Microsoft Word, Git, Visual Studio Code, Canva, Jupyter
- **Libraries:** matplotlib, numpy, Scikit-learn, pandas, yfinance
- **Athletics:** Boxing, Hockey, Soccer, Volleyball
- **Activities:** Tutoring, Mathematics Club, Undergraduate Research Club